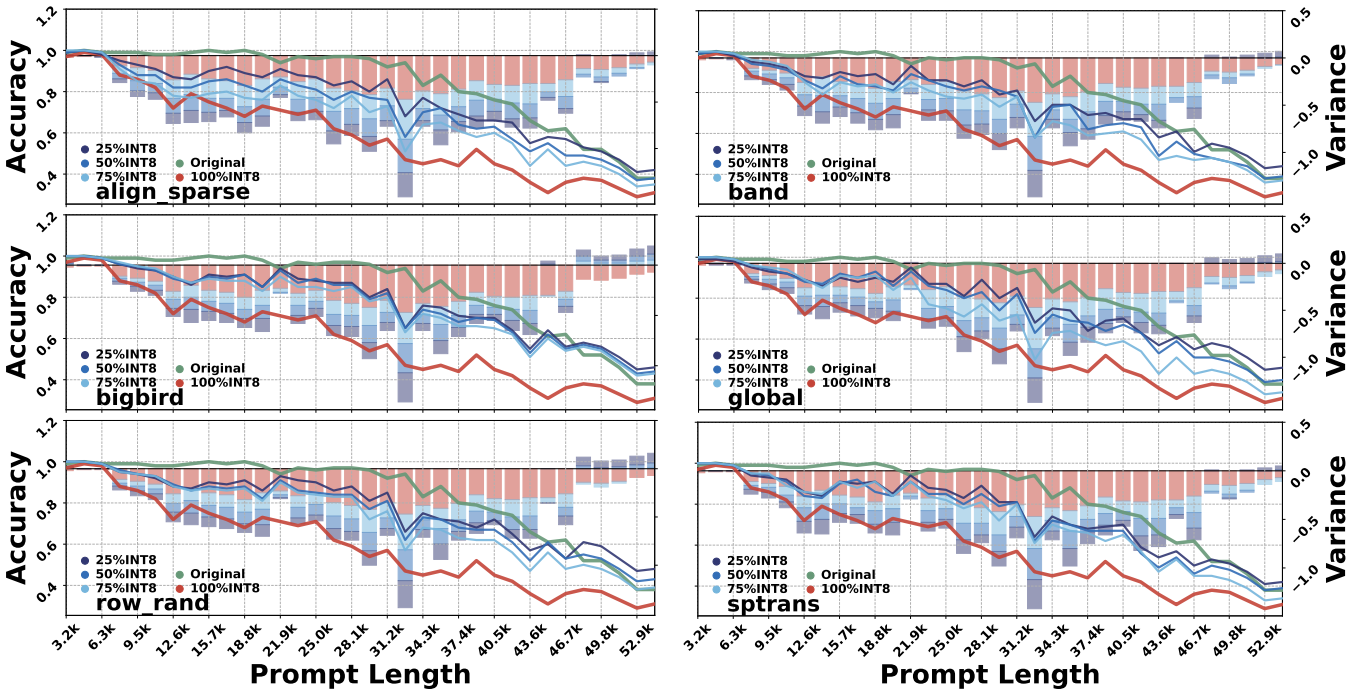


Figure 1: Line-level retrieval accuracy on the LongEval benchmark for Qwen 2.5 7B under different tile-group routing layouts, evaluated across prompt lengths from 3.1k to 38.7k tokens. Line plots (left y-axis) report exact-match retrieval accuracy, while bar plots (right y-axis, Δ accuracy vs. FP16) show differences relative to the FP16 attention baseline. Each panel corresponds to a routing layout with 25%, 50%, or 75% of tile groups routed to INT8. Colors indicate routing settings: green denotes the original FP16 attention baseline, red denotes One (all legal score-tile groups routed to INT8), and blue denotes mixed configurations with partially INT8-routed tile groups.

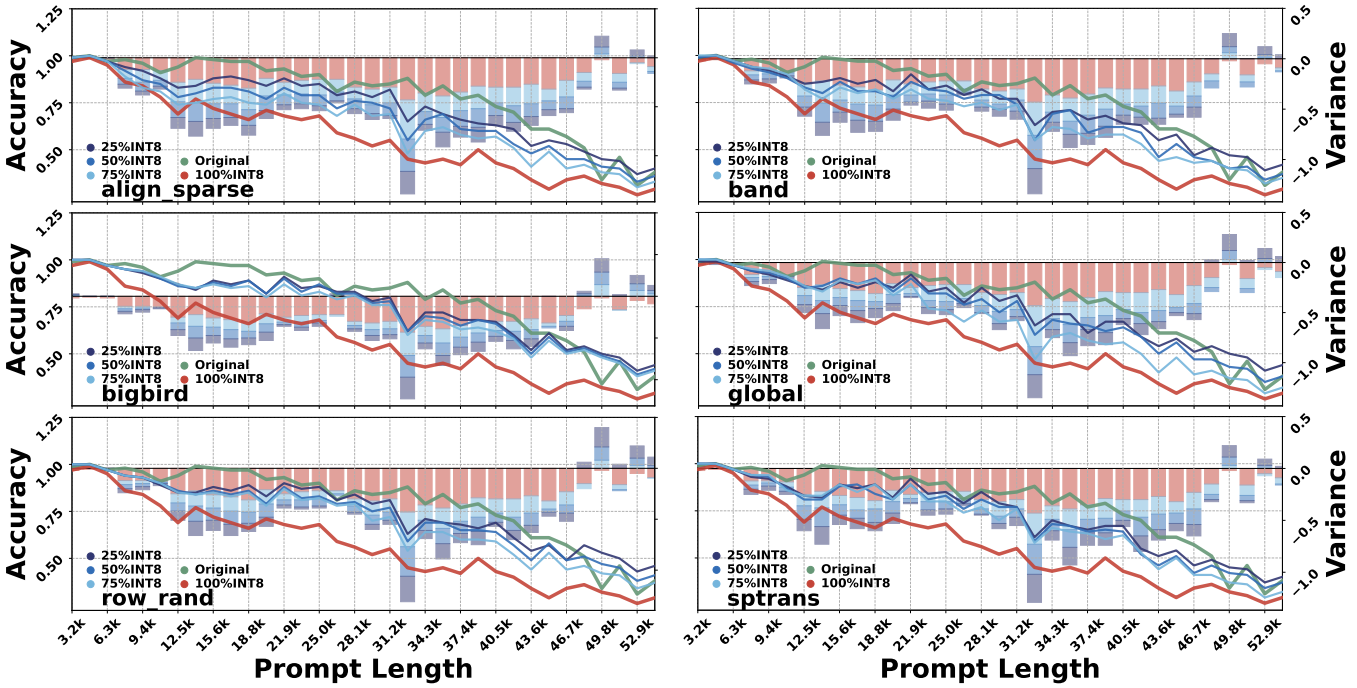


Figure 2: Line-level retrieval accuracy on the LongEval benchmark for Qwen 2 7B under different precision policy layouts, evaluated across prompt lengths from 3.1k to 38.7k tokens.

Dataset	Len	FP16	One	AlignSparse			Band			BigBird			Global			RowRand			SpTrans		
				25	50	75	25	50	75	25	50	75	25	50	75	25	50	75	25	50	75
①	16k	32.04	28.78	32.91	32.77	32.78	32.52	30.76	30.86	31.27	29.64	29.75	31.75	30.18	29.87	31.89	30.94	30.64	31.75	32.18	30.25
	32k	15.08	11.62	15.75	13.38	12.42	12.28	13.18	13.61	15.22	14.02	13.55	15.56	14.24	13.86	15.26	13.23	12.61	15.56	14.77	14.37
	64k	7.75	5.42	7.81	6.86	5.86	7.81	5.84	5.81	8.01	6.47	5.72	8.01	7.03	6.25	8.01	6.32	6.08	8.01	7.44	6.83
②	16k	18.49	15.62	18.95	17.08	16.83	19.16	16.57	17.23	18.61	16.98	16.62	18.64	17.62	16.88	18.39	16.58	16.56	18.64	18.08	16.78
	32k	15.12	12.02	15.18	14.34	13.16	15.48	13.59	13.09	15.14	13.53	13.28	15.27	14.78	13.59	15.58	14.15	13.09	15.27	14.69	14.18
	64k	11.84	7.79	12.15	10.00	9.12	12.30	10.17	10.10	12.04	10.28	10.03	12.06	10.96	10.28	12.19	10.39	10.13	12.06	11.34	11.04
③	16k	17.51	12.86	17.21	17.21	16.19	17.69	15.51	15.25	17.66	15.17	14.84	17.66	15.87	15.87	17.66	15.92	16.49	17.66	17.09	16.88
	32k	10.80	8.38	11.01	10.15	9.01	10.85	8.79	8.49	10.61	9.07	9.00	11.01	10.03	8.75	10.80	9.28	9.21	11.01	10.44	10.06
	64k	6.82	5.32	7.70	6.39	5.70	6.89	5.97	5.39	7.63	6.09	5.62	7.63	6.65	5.84	7.70	5.96	4.91	7.63	6.25	5.97
④	16k	12.33	8.51	12.43	11.77	9.79	11.66	9.70	6.91	12.38	10.40	10.30	12.31	11.33	10.52	12.02	10.27	10.53	12.31	11.74	11.03
	32k	6.73	3.22	6.12	5.76	3.88	7.04	4.84	4.90	6.52	5.18	4.45	7.08	6.10	4.73	6.66	5.19	4.56	7.08	6.51	5.95
	64k	1.84	0.74	1.99	1.87	1.02	0.84	0.86	0.86	1.91	1.35	1.13	1.87	1.59	1.05	1.99	1.89	1.24	1.97	1.37	1.22
⑤	16k	21.75	15.69	21.88	22.36	20.47	27.84	19.86	19.03	21.96	20.81	20.09	22.07	21.08	20.23	22.41	20.10	20.08	22.07	21.16	21.05
	32k	19.89	12.11	19.16	17.35	14.23	19.39	17.36	16.63	20.17	18.41	17.79	19.91	19.07	17.99	19.88	17.96	17.47	19.91	19.37	18.56
	64k	14.18	6.93	14.13	13.98	10.91	14.13	12.45	11.25	14.09	12.37	11.97	14.23	13.31	12.23	14.14	12.27	11.79	14.23	13.66	13.34
⑥	16k	18.24	12.72	18.16	16.12	13.13	18.50	16.65	15.53	17.57	16.45	15.96	17.53	16.97	15.75	18.12	16.27	16.28	17.53	16.99	17.46
	32k	14.25	9.12	13.98	12.13	11.43	14.44	12.64	11.40	13.73	12.73	12.29	13.63	13.03	12.21	14.19	12.35	12.17	13.63	13.18	13.00
	64k	11.08	7.54	11.06	9.11	8.28	11.22	9.23	8.07	11.05	9.60	9.04	11.31	10.09	9.56	11.08	9.18	8.73	11.31	10.65	10.38
⑦	16k	44.69	34.69	46.02	43.19	39.69	45.91	43.60	42.61	45.13	43.66	43.86	45.77	44.79	43.52	45.52	43.93	44.04	45.24	44.64	45.29
	32k	26.84	18.17	25.76	23.20	21.69	26.08	24.36	23.41	26.47	24.94	23.78	26.05	25.49	24.71	26.90	25.05	24.58	26.05	25.94	25.11
	64k	16.36	9.19	15.60	14.58	11.96	16.30	14.21	12.45	15.72	14.18	14.04	16.03	14.85	14.24	16.14	14.26	13.78	16.03	15.25	14.71
⑧	16k	28.88	22.37	28.26	26.21	25.80	28.80	27.06	25.79	28.95	27.57	26.33	29.10	28.13	27.15	28.17	26.31	26.42	29.10	28.31	27.23
	32k	18.33	13.66	18.03	15.99	15.98	17.84	16.20	15.13	18.17	16.70	16.88	18.26	17.35	16.44	18.25	16.55	16.51	18.26	17.65	17.41
	64k	15.75	12.81	16.08	15.91	14.98	16.18	14.13	13.18	15.93	14.22	13.88	15.94	14.94	14.13	15.85	14.77	14.41	15.94	15.32	15.39
⑨	16k	22.44	16.97	23.56	22.64	20.75	22.36	21.03	19.89	22.44	20.24	20.76	20.29	21.46	26.06	21.74	20.34	21.62	22.94	22.37	21.74
	32k	14.85	12.02	15.53	15.28	14.49	15.28	13.28	14.18	16.11	14.57	13.34	15.61	14.63	13.56	15.81	13.57	13.54	15.61	15.04	15.06
	64k	9.26	3.41	8.36	7.36	5.57	8.36	6.36	5.36	8.92	8.38	7.91	8.92	7.94	7.13	8.92	7.12	6.38	8.92	8.35	7.40

Table 1: LV-Eval long-context question answering for LLaMA 3.2 3B. Cells are color-coded by comparison to One and Baseline: red indicates performance below One, blue indicates performance between One and the Baseline, and green indicates performance above the Baseline.

Len	Metric	Torch	Flash	One	AlignSparse			Band			BigBird			Global			RowRand			SpTrans		
					25	50	75	25	50	75	25	50	75	25	50	75	25	50	75	25	50	75
1k	Thpt	11.14	17.45	32.27	29.22	32.02	33.48	27.06	29.30	34.56	26.92	31.48	32.37	26.71	29.59	32.03	28.66	32.09	34.95	27.66	32.11	33.50
	TOPS	41.70	65.31	120.80	109.38	119.86	125.29	101.31	109.71	129.42	100.78	117.85	121.17	99.98	110.75	119.92	107.25	120.12	130.80	103.50	120.20	125.38
2k	Thpt	7.78	16.48	32.06	29.50	32.41	33.84	26.77	28.89	34.30	26.57	28.83	32.71	26.45	29.29	31.74	26.34	31.35	34.88	27.11	31.46	33.92
	TOPS	30.50	64.64	125.70	115.70	127.08	132.67	104.99	113.30	134.48	104.23	113.05	128.31	103.69	114.83	124.46	103.28	122.91	136.77	106.32	123.38	132.99
4k	Thpt	**OOM**	14.33	29.80	27.71	30.35	31.62	25.65	28.59	32.81	24.69	26.84	27.48	24.50	27.19	29.35	26.97	30.06	32.89	27.14	30.59	31.80
	TOPS	**OOM**	61.31	127.48	118.53	129.85	135.26	109.76	122.32	140.39	105.63	114.83	117.56	104.82	116.35	125.56	115.37	128.62	140.73	116.09	130.85	136.03
8k	Thpt	**OOM**	**OOM**	27.41	23.17	25.39	26.44	23.23	25.03	28.19	23.02	25.17	25.84	22.84	25.39	27.47	22.78	25.33	27.74	22.81	25.69	26.61
	TOPS	**OOM**	**OOM**	136.76	115.64	126.72	131.96	115.89	124.89	140.73	114.92	125.58	128.90	113.98	126.74	137.09	113.72	126.35	138.42	113.83	128.21	132.84

Table 2: Throughput (Thpt, K tokens/s) and TOPS on LLaMA 3.2 3B-Instruct across sequence lengths. All methods are evaluated on the same A100 40GB hardware with batch size 8, 3 warmup iterations, and 5 measurement iterations under the same model wrapper. One denotes all legal score-tile groups routed to INT8. OOM entries are shown in gray and excluded from relative color comparisons.

L	Max	Mean	L	Max	Mean	L	Max	Mean	L	Max	Mean
0	0.00	0.00	12	5.63×10^{-2}	1.07×10^{-4}	24	5.82×10^{-2}	1.18×10^{-4}	36	5.97×10^{-2}	1.29×10^{-4}
1	3.23×10^{-3}	2.30×10^{-5}	13	5.63×10^{-2}	1.08×10^{-4}	25	5.82×10^{-2}	1.19×10^{-4}	37	5.95×10^{-2}	1.31×10^{-4}
2	6.38×10^{-3}	3.50×10^{-5}	14	5.62×10^{-2}	1.08×10^{-4}	26	5.88×10^{-2}	1.19×10^{-4}	38	5.92×10^{-2}	1.34×10^{-4}
3	9.78×10^{-3}	5.82×10^{-5}	15	5.62×10^{-2}	1.09×10^{-4}	27	5.90×10^{-2}	1.20×10^{-4}	39	5.88×10^{-2}	1.35×10^{-4}
4	5.10×10^{-2}	7.92×10^{-5}	16	5.62×10^{-2}	1.09×10^{-4}	28	5.94×10^{-2}	1.21×10^{-4}	40	5.82×10^{-2}	1.37×10^{-4}
5	5.43×10^{-2}	8.40×10^{-5}	17	5.63×10^{-2}	1.10×10^{-4}	29	5.96×10^{-2}	1.21×10^{-4}	41	5.76×10^{-2}	1.39×10^{-4}
6	5.64×10^{-2}	8.95×10^{-5}	18	5.62×10^{-2}	1.10×10^{-4}	30	5.96×10^{-2}	1.22×10^{-4}	42	5.74×10^{-2}	1.41×10^{-4}
7	5.65×10^{-2}	9.94×10^{-5}	19	5.62×10^{-2}	1.11×10^{-4}	31	5.98×10^{-2}	1.23×10^{-4}	43	5.71×10^{-2}	1.44×10^{-4}
8	5.64×10^{-2}	1.02×10^{-4}	20	5.62×10^{-2}	1.12×10^{-4}	32	5.98×10^{-2}	1.24×10^{-4}	44	3.65×10^{-2}	1.37×10^{-4}
9	5.64×10^{-2}	1.05×10^{-4}	21	5.62×10^{-2}	1.12×10^{-4}	33	5.98×10^{-2}	1.25×10^{-4}	45	2.96×10^{-2}	1.39×10^{-4}
10	5.64×10^{-2}	1.06×10^{-4}	22	5.78×10^{-2}	1.14×10^{-4}	34	5.98×10^{-2}	1.26×10^{-4}	46	2.98×10^{-2}	1.76×10^{-4}
11	5.64×10^{-2}	1.07×10^{-4}	23	5.79×10^{-2}	1.17×10^{-4}	35	5.98×10^{-2}	1.27×10^{-4}	47	8.93×10^{-4}	1.83×10^{-5}

Table 3: Layer-wise numerical differences between FP16 and FP32 accumulation on Qwen 2.5 14B. Max and Mean report the maximum and mean absolute differences at each layer.

Dataset	Len	One	SpTrans25	SpTrans50	SpTrans75	MInference	FlexPrefill	SageAttn
①	16k	31.60	36.00	34.30	33.40	29.00	26.13	32.60
	32k	14.70	20.00	19.00	18.50	15.00	11.89	16.90
	64k	6.60	9.70	8.80	8.20	6.40	8.98	9.10
②	16k	13.90	16.70	16.20	15.10	14.10	15.74	14.20
	32k	11.40	14.60	14.00	13.50	11.50	15.24	13.50
	64k	8.70	13.20	12.50	12.10	9.50	12.65	10.80
⑤	16k	12.24	17.10	16.50	17.54	14.90	17.09	15.80
	32k	9.30	15.30	14.90	14.40	12.60	16.08	13.80
	64k	5.80	11.70	11.10	10.80	9.00	10.29	11.40
⑥	16k	12.10	16.90	16.30	16.70	15.20	13.65	15.10
	32k	8.40	12.60	12.20	12.00	10.80	12.26	10.80
	64k	7.30	10.80	10.10	9.90	8.10	9.46	9.50
⑦	16k	25.17	34.19	35.77	36.00	28.50	29.07	31.40
	32k	15.70	22.90	22.60	22.10	18.10	20.71	21.00
	64k	9.00	15.90	15.50	15.00	11.40	14.53	13.70
⑧	16k	19.30	25.10	24.60	24.00	21.20	23.89	22.60
	32k	14.20	19.30	18.60	18.20	15.30	17.67	17.90
	64k	14.00	17.50	16.90	17.00	12.80	15.95	15.90
⑨	16k	25.30	34.30	33.50	32.30	28.70	22.74	31.40
	32k	20.10	25.30	24.60	24.50	19.30	15.98	23.50
	64k	6.90	18.40	17.20	15.30	13.00	15.22	16.10

Table 4: LV-Eval long-context question answering results on Qwen2-7B. We compare *TileMix* pure INT8 (One), *TileMix* SpTrans mixed-precision variants, and long-context attention baselines. SpTrans25, SpTrans50, and SpTrans75 denote configurations with 25%, 50%, and 75% INT8-routed tile groups. Datasets are denoted by the same indices defined in the experimental setup.

Dataset	Len	FP16	One	AlignSparse			Band			BigBird			Global			RowRand			SpTrans		
				25	50	75	25	50	75	25	50	75	25	50	75	25	50	75	25	50	75
①	16k	34.97	31.6	36.0	35.6	35.5	35.1	33.6	33.8	34.4	32.7	32.9	35.0	33.6	33.1	35.8	34.7	34.1	36.0	34.3	33.4
	32k	19.14	14.7	20.0	17.0	15.9	15.6	16.7	17.0	19.4	17.8	17.3	19.8	18.1	17.6	19.5	17.0	16.4	20.0	19.0	18.5
	64k	9.34	6.6	9.6	8.4	7.3	9.5	7.2	7.1	9.7	7.8	7.1	9.6	8.5	7.6	9.8	7.7	7.4	9.7	8.8	8.2
②	16k	16.39	13.9	16.8	15.4	15.1	17.1	15.0	15.5	16.4	15.0	14.8	16.6	15.7	15.0	16.4	14.9	14.9	16.7	16.2	15.1
	32k	14.42	11.4	14.3	13.4	12.6	14.6	12.9	12.4	14.5	13.0	12.6	14.6	14.0	12.9	14.8	13.4	12.7	14.6	14.0	13.5
	64k	12.97	8.7	13.3	11.0	10.3	13.4	11.2	11.0	13.2	11.4	10.9	13.3	12.0	11.2	13.5	11.6	11.2	13.2	12.5	12.1
③	16k	14.21	10.6	14.2	14.1	13.2	14.4	12.8	12.5	14.4	12.4	12.1	14.4	12.9	12.8	14.3	12.7	13.5	14.5	14.0	13.8
	32k	8.16	6.3	8.3	7.6	6.7	8.1	6.6	6.4	8.0	6.9	6.8	8.3	7.6	6.7	8.2	7.1	7.0	8.3	7.8	7.5
	64k	6.81	5.3	7.6	6.3	5.6	7.4	6.4	5.7	7.7	6.6	6.1	7.7	6.7	6.0	7.8	6.1	5.1	7.7	6.3	6.0
④	16k	23.28	16.0	23.3	22.1	18.6	21.9	18.2	13.3	23.2	19.7	19.4	23.0	21.2	19.8	22.6	19.3	19.9	23.2	22.1	21.3
	32k	13.73	6.6	13.1	12.3	8.3	14.4	9.9	10.0	13.4	10.7	9.1	14.6	12.6	9.7	13.8	10.7	9.4	14.6	13.5	12.3
	64k	6.91	2.9	7.4	7.0	3.8	3.2	3.3	3.3	7.2	5.1	4.3	7.0	6.0	3.9	7.4	7.0	4.7	7.3	5.2	4.6
⑤	16k	16.86	12.24	17.1	17.3	15.9	20.83	15.1	14.6	17.0	15.9	15.4	17.2	16.4	15.8	17.3	15.6	15.5	17.1	16.5	17.54
	32k	15.09	9.3	14.7	13.6	11.1	14.9	13.3	12.7	15.4	14.1	13.5	15.2	14.6	13.7	15.1	13.8	13.4	15.3	14.9	14.4
	64k	11.62	5.8	11.6	11.2	8.7	11.5	10.1	9.2	11.6	10.2	9.8	11.6	10.9	10.2	11.6	10.1	9.6	11.7	11.1	10.8
⑥	16k	17.36	12.1	17.2	15.3	12.5	17.7	16.0	14.9	16.8	15.7	15.2	16.6	16.1	15.1	17.2	15.6	15.7	16.9	16.3	16.7
	32k	13.03	8.4	12.7	11.0	10.2	13.3	11.6	10.5	12.6	11.6	11.2	12.4	11.8	11.1	13.1	11.5	11.3	12.6	12.2	12.0
	64k	10.73	7.3	10.8	9.1	8.2	11.0	9.3	8.2	10.7	9.2	8.7	10.9	9.8	9.3	10.7	9.3	8.9	10.8	10.1	9.9
⑦	16k	33.30	25.17	33.7	31.7	29.1	33.8	32.2	31.5	33.4	32.2	32.4	33.7	33.0	32.1	33.6	32.4	32.6	34.19	35.77	36.00
	32k	23.30	15.7	22.6	20.5	19.0	22.8	21.3	20.6	23.2	21.9	20.9	22.9	22.4	21.6	23.6	22.0	21.5	22.9	22.6	22.1
	64k	15.95	9.0	16.0	14.8	12.1	15.9	13.9	12.5	15.7	14.2	14.1	16.0	14.9	14.3	16.1	14.4	13.9	15.9	15.5	15.0
⑧	16k	24.88	19.3	24.4	22.9	22.5	25.0	23.3	22.2	25.2	23.9	23.0	25.3	24.5	23.7	24.7	23.1	23.3	25.1	24.6	24.0
	32k	19.22	14.2	18.8	16.7	16.5	18.8	17.1	16.1	19.1	17.6	17.3	19.3	18.4	17.5	19.2	17.5	17.3	19.3	18.6	18.2
	64k	17.20	14.0	17.7	17.2	16.3	17.6	15.4	14.6	17.4	15.7	15.4	17.5	16.4	15.6	17.3	16.2	15.9	17.5	16.9	17.0
⑨	16k	33.70	25.3	35.3	33.9	31.2	33.5	31.5	29.8	33.7	30.3	31.0	33.9	32.4	31.2	33.6	31.0	32.1	34.3	33.5	32.3
	32k	24.82	20.1	25.6	25.0	23.7	25.2	22.0	23.6	26.2	23.7	21.8	25.4	23.8	22.1	25.9	22.3	22.2	25.3	24.6	24.5
	64k	18.51	6.9	17.1	15.3	11.6	17.1	13.1	11.1	18.3	17.2	16.3	18.3	16.4	14.9	18.2	14.6	13.1	18.4	17.2	15.3

Table 5: LV-Eval for Qwen 2 7B. Cells are color-coded by comparison to One and Baseline: red indicates performance below One, blue indicates performance between One and the Baseline, and green indicates performance above the Baseline.

Len	Metric	Torch	Flash	One	AlignSparse			Band			BigBird			Global			RowRand			SpTrans		
					25	50	75	25	50	75	25	50	75	25	50	75	25	50	75	25	50	75
1k	Thpt	7.16	8.21	19.78	16.20	19.82	18.04	16.12	17.93	17.95	16.23	18.09	18.07	16.20	18.04	20.56	16.25	18.02	17.98	16.25	20.26	20.14
	TOPS	56.51	64.81	156.18	127.94	156.43	142.38	127.26	141.56	141.74	128.19	142.87	142.69	127.95	142.38	162.36	128.23	142.28	141.99	128.34	159.91	159.07
2k	Thpt	5.35	7.89	19.56	15.99	19.56	17.77	15.89	18.52	19.29	16.04	17.86	17.78	16.02	17.78	20.29	16.01	17.77	17.77	16.01	19.86	19.76
	TOPS	43.34	63.93	158.52	129.50	158.47	143.92	128.76	150.08	156.33	129.92	144.62	144.07	129.79	144.14	164.38	129.65	144.00	144.00	129.76	160.98	160.15
4k	Thpt	**OOM**	7.22	18.49	15.17	16.80	16.80	15.17	17.62	18.31	15.26	16.91	16.80	15.25	16.91	19.23	15.22	16.85	16.82	15.07	18.76	18.66
	TOPS	**OOM**	61.48	157.51	129.19	143.03	143.12	129.21	150.04	156.04	130.00	144.09	143.19	129.94	144.05	163.77	129.58	143.53	143.21	128.32	159.69	158.85

Table 6: Throughput (Thpt, K tokens/s) and TOPS on Qwen 2 7B across sequence lengths.

Dataset	Len	FP16	One	AlignSparse			Band			BigBird			Global			RowRand			SpTrans		
				25	50	75	25	50	75	25	50	75	25	50	75	25	50	75	25	50	75
①	16k	33.88	28.9	34.7	34.4	34.5	33.9	32.0	32.1	33.2	31.1	31.3	33.7	32.0	31.7	34.8	33.7	33.2	34.9	33.1	32.2
	32k	17.61	13.6	18.3	15.6	14.6	14.4	15.3	15.7	17.8	16.4	15.9	18.2	16.6	16.1	17.8	15.5	14.9	18.3	17.4	16.9
	64k	9.22	6.5	9.3	8.1	7.0	9.3	7.0	6.9	9.5	7.6	6.8	9.4	8.3	7.4	9.5	7.4	7.1	9.4	8.5	7.9
②	16k	17.37	14.8	17.8	16.2	15.9	18.1	15.8	16.4	17.4	15.9	15.6	17.6	16.7	15.9	17.4	15.8	15.7	17.6	17.2	16.0
	32k	15.89	12.6	15.8	14.8	13.9	16.0	14.1	13.6	15.9	14.2	13.8	16.0	15.4	14.2	16.2	14.7	13.9	16.0	15.3	14.8
	64k	12.12	8.1	12.4	10.2	9.5	12.5	10.4	10.2	12.3	10.6	10.1	12.4	11.2	10.5	12.6	10.8	10.4	12.3	11.6	11.2
③	16k	19.13	14.2	18.9	18.8	17.9	19.2	17.1	16.7	19.2	16.7	16.4	19.2	17.3	17.1	19.1	17.0	18.0	19.3	18.7	18.2
	32k	8.72	6.8	8.9	8.1	7.3	8.7	7.1	6.9	8.6	7.4	7.3	8.9	8.0	7.1	8.8	7.6	7.5	8.9	8.3	8.0
	64k	8.09	6.3	9.0	7.5	6.7	8.8	7.7	6.9	9.1	7.8	7.2	9.1	7.9	7.1	9.2	7.2	6.0	9.0	7.4	7.1
④	16k	23.46	16.23	23.84	21.48	18.31	22.47	17.66	12.93	23.57	20.24	18.82	23.06	20.78	19.18	22.74	19.61	19.37	23.51	22.26	20.76
	32k	16.05	7.7	15.3	14.4	9.7	16.9	11.6	11.8	15.7	12.5	10.7	17.1	14.8	11.4	16.1	12.5	11.0	17.1	15.8	14.4
	64k	5.58	2.3	6.0	5.7	3.1	2.6	2.7	2.7	5.8	4.1	3.4	5.7	4.9	3.2	6.0	5.7	3.8	5.9	4.2	3.7
⑤	16k	21.78	15.52	21.8	22.1	20.3	26.82	19.1	18.5	21.7	20.4	19.8	21.9	20.9	20.1	22.0	20.2	20.0	21.8	21.0	21.30
	32k	18.85	11.6	18.3	16.9	13.8	18.7	16.7	16.0	19.4	17.8	17.0	19.2	18.4	17.3	19.1	17.4	16.9	19.3	18.8	18.1
	64k	12.67	6.3	12.6	12.1	9.4	12.5	11.0	10.0	12.6	11.1	10.7	12.7	11.9	11.1	12.6	11.0	10.5	12.7	12.1	11.8
⑥	16k	21.27	14.8	21.1	18.7	15.2	21.6	19.5	18.2	20.5	19.2	18.6	20.3	19.7	18.4	21.1	19.1	19.2	20.6	19.8	20.3
	32k	16.56	10.7	16.2	14.1	13.1	16.9	14.8	13.4	16.1	14.9	14.4	15.9	15.2	14.3	16.7	14.6	14.4	16.1	15.6	15.4
	64k	12.90	8.8	12.9	10.9	9.8	13.2	11.1	9.8	12.8	11.0	10.4	13.1	11.8	11.2	12.9	11.2	10.7	13.0	12.2	11.9
⑦	16k	41.68	30.52	41.9	39.3	36.1	41.8	39.9	39.0	41.3	39.9	40.1	41.7	40.9	39.7	41.6	40.1	40.3	41.45	39.81	40.82
	32k	26.25	17.7	25.4	23.0	21.3	25.7	24.0	23.2	26.1	24.6	23.5	25.8	25.2	24.3	26.6	24.8	24.2	25.9	25.6	25.0
	64k	14.08	7.9	14.1	13.0	10.6	14.0	12.2	11.0	13.8	12.5	12.4	14.1	13.1	12.6	14.2	12.7	12.2	14.0	13.6	13.1
⑧	16k	26.81	20.8	26.3	24.6	24.2	26.9	25.1	23.9	27.1	25.7	24.7	27.2	26.3	25.5	26.6	24.9	25.1	27.0	26.4	25.8
	32k	18.63	13.8	18.2	16.2	16.0	18.2	16.6	15.6	18.5	17.1	16.8	18.7	17.8	16.9	18.6	16.9	16.7	18.7	18.0	17.6
	64k	16.42	13.4	16.9	16.4	15.5	16.8	14.7	13.9	16.6	15.0	14.7	16.7	15.7	14.9	16.5	15.4	15.1	16.7	16.1	16.2
⑨	16k	34.06	25.45	34.84	33.26	30.58	33.91	30.84	29.43	34.01	30.77	31.06	33.88	32.66	30.47	33.16	31.27	31.54	34.44	32.76	32.63
	32k	22.36	18.1	23.1	22.6	21.4	22.7	19.8	21.2	23.6	21.3	19.6	22.9	21.4	19.9	23.3	20.1	20.0	22.8	22.1	22.0
	64k	14.06	5.2	12.9	11.5	8.8	13.0	9.9	8.4	13.9	13.0	12.3	13.9	12.4	11.2	13.8	11.0	9.9	14.0	13.1	11.6

Table 7: LV-Eval long-context question answering for Qwen 2.5 7B under different precision layouts and INT8 ratios (25/50/75%). Results are reported for FP16 attention (Baseline), One (all legal score-tile groups routed to INT8), and mixed-precision configurations. Cells are color-coded by comparison to One and Baseline: red indicates performance below One, blue indicates performance between One and the Baseline, and green indicates performance above the Baseline.

Len	Metric	Torch	Flash	One	AlignSparse			Band			BigBird			Global			RowRand			SpTrans		
					25	50	75	25	50	75	25	50	75	25	50	75	25	50	75	25	50	75
1k	Thpt	6.98	7.98	19.27	15.10	16.12	18.06	14.28	16.98	17.59	15.03	16.47	18.18	14.74	16.53	17.93	14.90	15.87	17.96	15.09	16.05	17.76
	TOPS	55.11	63.01	152.10	119.20	127.24	142.63	112.75	134.02	138.79	118.71	130.01	143.58	116.37	130.48	141.49	117.63	125.32	141.87	119.14	126.73	140.17
2k	Thpt	5.21	7.72	19.13	14.95	15.93	17.84	14.18	16.80	17.41	14.88	16.23	17.94	14.63	16.35	17.76	14.71	15.71	17.76	14.89	15.84	17.52
	TOPS	42.23	62.54	155.00	121.07	129.05	144.61	114.88	136.17	140.99	120.58	131.59	145.36	118.51	132.50	143.85	119.23	127.32	143.91	120.61	128.32	141.98
4k	Thpt	**OOM**	7.09	18.03	14.09	14.95	16.74	13.50	15.98	16.43	14.12	15.37	16.84	13.89	15.43	16.65	13.89	14.75	16.65	14.03	14.97	16.49
	TOPS	**OOM**	60.41	153.54	120.09	127.37	142.70	115.00	136.10	139.94	120.31	130.89	143.35	118.25	131.40	141.80	118.37	125.64	141.89	119.51	127.56	140.44

Table 8: Throughput (Thpt, K tokens/s) and TOPS on Qwen 2.5 7B across sequence lengths.

Line	Len	FP16	INT8	AlignSparse			Band			BigBird			Global			RowRand			SpTrans		
				25	50	75	25	50	75	25	50	75	25	50	75	25	50	75	25	50	75
200	4776.33	0.99	0.97	0.99	0.99	1.00	0.99	0.99	1.00	1.00	1.00	1.00	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
300	7081.42	1.00	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
400	9384.62	0.99	0.98	0.98	0.99	0.99	0.98	0.99	0.99	0.99	0.99	0.99	0.98	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99
500	11691.21	0.85	0.76	0.82	0.80	0.78	0.82	0.81	0.82	0.82	0.82	0.83	0.82	0.82	0.82	0.82	0.82	0.82	0.81	0.82	0.82
600	13991.58	0.73	0.63	0.69	0.65	0.63	0.69	0.69	0.68	0.69	0.70	0.70	0.69	0.69	0.70	0.69	0.69	0.69	0.69	0.69	0.67
700	16294.20	0.52	0.44	0.48	0.47	0.45	0.49	0.48	0.46	0.49	0.49	0.50	0.49	0.49	0.50	0.49	0.49	0.48	0.49	0.48	0.47

Table 9: Line-level retrieval accuracy on the LongEval benchmark for Vicuna 7B under different precision layouts inspired by sparse-attention patterns. The task requires the model to scan long input sequences and exactly retrieve the content associated with a queried line identifier. Accuracy is measured as exact-match retrieval rate, reported across increasing input lengths. Results compare full FP16 attention, pure INT8 attention, and mixed-precision configurations with varying precision layouts and INT8 ratios, illustrating how different precision-allocation patterns affect robustness under low-precision execution.

Len	Metric	Torch	Flash	One	AlignSparse			Band			BigBird			Global			RowRand			SpTrans		
					25	50	75	25	50	75	25	50	75	25	50	75	25	50	75	25	50	75
1k	Thpt	6.52	8.40	18.13	15.54	16.71	18.02	15.38	16.71	18.02	15.32	16.48	17.88	15.23	16.38	18.36	15.08	16.54	17.83	15.01	16.15	17.52
	TOPS	44.86	66.30	124.65	106.85	114.97	123.98	105.80	114.97	123.98	105.39	113.33	122.93	104.72	112.67	126.36	103.68	113.78	122.71	103.28	111.07	120.47
2k	Thpt	4.60	8.07	17.35	14.91	16.05	17.30	14.76	16.05	17.30	14.66	15.88	17.11	14.61	15.73	17.64	14.46	15.89	17.13	14.37	15.56	16.77
	TOPS	32.88	65.40	124.02	106.64	114.82	123.67	105.58	114.82	123.67	104.81	113.53	122.37	104.51	112.53	126.05	103.47	113.64	122.40	102.72	111.26	119.93
4k	Thpt	**OOM**	7.39	15.80	13.78	14.80	15.92	13.64	14.80	15.92	13.51	14.67	15.66	13.51	14.51	16.22	13.37	14.65	15.75	13.24	14.38	15.35
	TOPS	**OOM**	62.89	121.58	106.03	113.93	122.50	104.98	113.93	122.50	103.93	112.87	120.48	103.91	111.65	124.85	102.88	112.76	121.24	101.85	110.62	118.07

Table 10: Throughput (Thpt, K tokens/s) and TOPS on Vicuna 7B across sequence lengths.